

Javad Komijani

Curriculum Vitae

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Professional Summary

Highly analytical and detail-oriented researcher with an extensive background in mathematical methods, numerical simulations, theoretical modeling, statistical analysis, signal processing, and machine learning. Holds an MSc in Electrical Engineering and a PhD in Physics. Passionate about applying data science and machine learning to solve complex, real-world problems.

Education

2015 **PhD in Physics**, *Washington University in St. Louis*

Thesis: *Topics in Lattice Gauge Theory and Theoretical Physics*

- Developed theoretical models based on effective field theories
- Applied Bayesian inference to lattice-QCD data
- Estimated parameters of the Standard Model
- Introduced new concepts in mathematical physics

2009 **MSc in Electrical Engineering**, *University of Tehran*

- Focus: electromagnetic waves and fields, signal processing, statistical analysis

2006 **BSc in Electrical Engineering**, *University of Tehran*

Technical Skills

Programming & Tools Python, Cython, C/C++, bash, MATLAB, Mathematica, PyTorch, Pandas, NumPy, SciPy, Scikit-learn, SQL, git

ML & AI Generative Models, Diffusion Models, Normalizing Flows

Statistics Bayesian inference, Monte Carlo methods, statistical analysis, optimization

Computation Numerical Simulations, High-Performance Computing (HPC)

Selected Projects

Normflow Normalizing Flows in PyTorch; supporting lattice gauge theory.

torch_linagl Linear Algebra Extension for PyTorch; supporting automatic differentiation for eigenvalue and singular value decompositions of unitary matrices.

torch_solve PyTorch Extension for Solving ODE; supporting automatic differentiation using the adjoint sensitivity method for unitary matrices.

Diffusion Diffusion Models in PyTorch; supporting unitary matrices.

interTwin Contributed the normflow package to the interTwin physics thematic module—integrated normalizing flows into the interTwin Digital Twin Engine for lattice QCD simulations.

Quark Masses Precisely calculated the up-, down-, strange-, charm-, and bottom-quark masses using the MILC highly improved staggered-quark ensembles.

MRS Mass Defined a novel quark mass scheme, the minimal-renormalon-subtracted (MRS) mass, which is free from the shortcomings of other quark masses and provides a more robust foundation for precise quark mass determination.

Eigenvalues Generalized eigenvalue problems to non-linear cases using properties of solutions to the Schrödinger equation.

Experience

- 2021–Present **Senior Postdoctoral Researcher, ETH Zurich**
- Developed ML algorithms to generate lattice field configurations
 - Worked extensively with HPC systems for lattice-QCD simulations
 - Collaborated internationally on computational/theoretical projects
 - Supervised PhD/MSc students and taught Quantum Mechanics
- 2019–2020 **Postdoctoral Researcher, University of Tehran**
- Applied lattice-QCD simulations and Bayesian analysis to determine Standard Model parameters
- 2017–2018 **Postdoctoral Researcher, University of Glasgow**
- Focused on determination of various form factors and quark condensates, using lattice-QCD Monte-Carlo simulations and Bayesian analysis
- 2015–2017 **Postdoctoral Researcher, Technical University of Munich**
- Focused on determination of quark masses and various decay constants, using lattice-QCD Monte-Carlo simulations and Bayesian analysis

Language Skills

English Fluent
Persian Mother tongue
German Very little

Selected Publications

- 2025 O. Vega, J. Komijani, A. El-Khadra, M. K. Marinkovic, *Group-Equivariant Diffusion Models for Lattice Field Theory*, arXiv:2510.26081
- 2025 J. Komijani, M. K. Marinkovic, *Normalizing flows for $SU(N)$ gauge theories employing singular value decomposition*, arXiv:2501.18288
- 2023 J. Komijani, M. K. Marinkovic, *Generative models for scalar field theories: how to deal with poor scaling?*, arXiv:2301.01504
- 2021 J. Komijani, *First-order nonlinear eigenvalue problems involving functions of a general oscillatory behavior*, J. Phys. A 54, 465202
- 2020 J. Komijani et al., *Strong coupling constant and quark masses from lattice QCD*, PPNP 113, 103788
- 2019 C.M. Bender et al., *Nonlinear eigenvalue problems for generalized Painlevé equations*, J. Phys. A 52, 315202
- 2018 A. Bazavov et al., *Up, down, strange, charm, and bottom quark masses from four-flavor lattice QCD*, PRD 98, 054517 (2018)
- 2017 J. Komijani, *Discussion on leading renormalon in the pole mass*, JHEP 1708, 062
- 2015 C.M. Bender, J. Komijani, *Painlevé Transcendents and PT-Symmetric Hamiltonians*, J. Phys. A 48, 475202